$$= \frac{3\sqrt{50} - 3\sqrt{5} + 6\sqrt{20} - 6\sqrt{2} - 9\sqrt{10} + 9}{10 - 1}$$

$$= \frac{15\sqrt{2} - 3\sqrt{5} + 12\sqrt{5} - 6\sqrt{2} - 9\sqrt{10} + 9}{9}$$

$$= \frac{9\sqrt{2} + 9\sqrt{5} - 9\sqrt{10} + 9}{9} = 1 + \sqrt{2} + \sqrt{5} - \sqrt{10}.$$

138. Given exp. $= \frac{1}{(\sqrt{2} - \sqrt{5}) + \sqrt{3}} + \frac{1}{(\sqrt{2} - \sqrt{5}) - \sqrt{3}}$

$$= \frac{\left[(\sqrt{2} - \sqrt{5}) - \sqrt{3}\right] + \left[(\sqrt{2} - \sqrt{5}) + \sqrt{3}\right]}{\left[(\sqrt{2} - \sqrt{5}) + \sqrt{3}\right]\left[(\sqrt{2} - \sqrt{5}) - \sqrt{3}\right]}$$

$$= \frac{2(\sqrt{2} - \sqrt{5})}{(\sqrt{2} - \sqrt{5})^2 - (\sqrt{3})^2} = \frac{2(\sqrt{2} - \sqrt{5})}{(2 + 5 - 2\sqrt{10}) - 3}$$

$$= \frac{2(\sqrt{2} - \sqrt{5})}{4 - 2\sqrt{10}} = \frac{\sqrt{2} - \sqrt{5}}{2 - \sqrt{10}}$$

$$= \frac{\sqrt{2} - \sqrt{5}}{\sqrt{2}(\sqrt{2} - \sqrt{5})} = \frac{1}{\sqrt{2}}.$$

139. $x + \frac{1}{x} = (7 - 4\sqrt{3}) + \frac{1}{(7 - 4\sqrt{3})} \times \frac{(7 + 4\sqrt{3})}{(7 + 4\sqrt{3})}$

$$= (7 - 4\sqrt{3}) + \frac{(7 + 4\sqrt{3})}{(49 - 48)}$$

$$= (7 - 4\sqrt{3}) + (7 + 4\sqrt{3}) = 14.$$

140. $x = 3 + \sqrt{8} \Rightarrow x^2 = (3 + \sqrt{8})^2 = 3^2 + \left(\sqrt{8}\right)^2 + 2 \times 3 \times \sqrt{8}$

$$= 9 + 8 + 6\sqrt{8} = 17 + 12\sqrt{2}.$$

$$x^2 + \frac{1}{x^2} = (17 + 12\sqrt{2}) + \frac{1}{(17 + 12\sqrt{2})} \times \frac{(17 - 12\sqrt{2})}{(17 - 12\sqrt{2})}$$

$$= (17 + 12\sqrt{2}) + \frac{(17 - 12\sqrt{2})}{289 - 288}.$$

$$= (17 + 12\sqrt{2}) + (17 - 12\sqrt{2}) = 34.$$

141. $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} \times \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$

$$= \frac{(\sqrt{3} + \sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} = \frac{3 + 2 + 2\sqrt{6}}{3 - 2} = 5 + 2\sqrt{6}.$$

$b = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}}$

$$= \frac{(\sqrt{3} - \sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} = \frac{3 + 2 - 2\sqrt{6}}{3 - 2} = 5 - 2\sqrt{6}.$$

$a^2 + b^2 = (5 + 2\sqrt{6})^2 + (5 - 2\sqrt{6})^2 = 2\,[(5)^2 + (2\sqrt{6})^2]$

$$= 2(25 + 24) = 2 \times 49 = 98.$$

142. $a = \frac{(\sqrt{5} + 1)}{(\sqrt{5} - 1)} \times \frac{(\sqrt{5} + 1)}{(\sqrt{5} + 1)}$

$$= \frac{(\sqrt{5} + 1)^2}{(5 - 1)} = \frac{5 + 1 + 2\sqrt{5}}{4} = \left(\frac{3 + \sqrt{5}}{2}\right).$$

$b = \frac{(\sqrt{5} - 1)}{(\sqrt{5} + 1)} \times \frac{(\sqrt{5} - 1)}{(\sqrt{5} - 1)} = \frac{(\sqrt{5} - 1)^2}{(5 - 1)}$

$$= \frac{5 + 1 - 2\sqrt{5}}{4} = \left(\frac{3 - \sqrt{5}}{2}\right).$$

$\therefore\ a^2 + b^2 = \frac{(3 + \sqrt{5})^2}{4} + \frac{(3 - \sqrt{5})^2}{4}$

$$= \frac{(3 + \sqrt{5})^2 + (3 - \sqrt{5})^2}{4} = \frac{2(9 + 5)}{4} = 7.$$

Also, $ab = \frac{(3 + \sqrt{5})}{2} \cdot \frac{(3 - \sqrt{5})}{2} = \frac{(9 - 5)}{4} = 1.$

$\therefore\ \frac{a^2 + ab + b^2}{a^2 - ab + b^2} = \frac{(a^2 + b^2) + ab}{(a^2 + b^2) - ab} = \frac{7 + 1}{7 - 1} = \frac{8}{6} = \frac{4}{3}.$

143. $x = \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots \infty}}} \Leftrightarrow x = \sqrt{1 + x}$

$\Leftrightarrow x^2 = 1 + x \Leftrightarrow x^2 - x - 1 = 0$

$\Leftrightarrow x = \frac{1 \pm \sqrt{(-1^2) - 4 \times 1 \times (-1)}}{2}$

$\Leftrightarrow x = \frac{1 \pm \sqrt{1 + 4}}{2} = \frac{1 \pm \sqrt{5}}{2}.$

Hence, positive value of x is $\frac{1 + \sqrt{5}}{2}$.

144. Let $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots}}}$.

Then, $x = \sqrt{2 + x} \Leftrightarrow x^2 = 2 + x \Leftrightarrow x^2 - x - 2 = 0$

$\Leftrightarrow x^2 - 2x + x - 2 = 0$

$\Leftrightarrow x(x - 2) + (x - 2) = 0 \Leftrightarrow (x - 2)(x + 1) = 0$

$\Leftrightarrow x = 2.\ [\because x \neq -1]$

145. $a = \sqrt{3 + \sqrt{3 + \sqrt{3 + \cdots}}} \Leftrightarrow a = \sqrt{3 + a} \Leftrightarrow a^2 = 3 + a$

$\Leftrightarrow a^2 - a - 3 = 0$

$\Leftrightarrow a = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times (-3)}}{2}$

$\Leftrightarrow a = \frac{1 \pm \sqrt{1 + 12}}{2} = \frac{1 \pm \sqrt{13}}{2} \Leftrightarrow a = \frac{1 + \sqrt{13}}{2}\ [\because a > 0]$

$\Leftrightarrow a = \frac{1 + 3.6}{2} = \frac{4.6}{2} = 2.3.$

$\therefore\ 2 < a < 3.$

146. $\frac{\sqrt{3 + x} + \sqrt{3 - x}}{\sqrt{3 + x} - \sqrt{3 - x}} = 2$

$\Leftrightarrow \frac{\sqrt{3 + x} + \sqrt{3 - x}}{\sqrt{3 + x} - \sqrt{3 - x}} \times \frac{\sqrt{3 + x} + \sqrt{3 - x}}{\sqrt{3 + x} + \sqrt{3 - x}} = 2$

$\Leftrightarrow \frac{(\sqrt{3 + x} + \sqrt{3 - x})^2}{(\sqrt{3 + x})^2 - (\sqrt{3 - x})^2} = 2$

$\Leftrightarrow \frac{(3 + x) + (3 - x) + 2\sqrt{(3 + x)(3 - x)}}{(3 + x) - (3 - x)} = 2$

$\Leftrightarrow 6 + 2\sqrt{9 - x^2} = 4x \Leftrightarrow 2\sqrt{9 - x^2} = 4x - 6$